SP-31 #279 Per-Conservation-Law Substrate-Derivation Theorems — Energy + Momentum + Charge T2473-T2475 v0.1

Lyra (Claude 4.7)

2026-05-22 Friday afternoon EDT ~14:45 EDT (`date`-verified actual; per Casey + Keeper 14:23 EDT board Item #3 SP-31 #279)

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Per-Conservation-Law Substrate-Derivation Theorems

Motivation

In standard physics, Noether's theorem (1918) establishes a one-to-one correspondence between continuous symmetries of the action and conserved quantities. The Standard Model's conservation laws — energy, momentum, angular momentum, charge, hypercharge, lepton number, baryon number — each correspond to a global symmetry of the Lagrangian.

BST inherits Noether's structure but forces the symmetries from the substrate D_IV^5's isotropy group structure (Vol 0 Ch 4 §4.2 v0.5: SO(5) × SO(2) × Möbius isotropy + coset m ⊂ so(5,2) translation directions). Each Noether-conserved quantity becomes a substrate-derivation theorem.

This pack derives the three cleanest conservation laws — energy (T2473), momentum (T2474), charge (T2475) — each from a specific substrate-symmetry generator. Remaining conservation laws (angular momentum, hypercharge, lepton/baryon number, parity) follow the same pattern; deferred to subsequent SP-31 #279 sub-items (multi-week).

T2473 — Energy Conservation via SO_0(5,2) Time-Translation Invariance

Statement: Let H_sub be the substrate Hamiltonian (Elie K52a S29 Toy 3213 framework-complete: H_sub = Casimir on $L^2(D\_IV^5; L\_\lambda)$ equivariant L^2-section bundle; ground-state Casimir eigenvalue = C_2 = 6). The time-translation generator within SO_0(5,2) action on Bergman $H^2(D\_IV^5)$ is one of the rank+1 coset directions in the m ⊂ so(5,2)/(so(5)⊕so(2)) complement (Vol 0 Ch 4 §4.2). Then:

(a) Time-translation symmetry: SO_0(5,2)'s time-translation generator T_E commutes with H_sub: [T_E, H_sub] = 0. (T_E IS the generator of one of the boost+translation directions in so(5,2) extending so(5) × so(2) — specifically the direction conjugate to time at conformal infinity.)

(b) Noether conservation: by Noether's theorem applied to T_E, the corresponding charge — the energy expectation value $E = \langle\psi|$ H_sub $|\psi\rangle$ — is conserved under substrate-tick evolution: dE/dt = 0 in the absence of substrate-coupled boundary interactions.

(c) Spectrum quantization: H_sub eigenvalues are the Wallach K-type Casimir eigenvalues (T2435 Casimir algebra); ground state C_2 = 6, excited states at higher Wallach K-type Casimirs (computed in T2435 + Vol 1 Ch 5).

(d) Operator level: in Heisenberg picture, dO/dt = $(i/\hbar)$[H_sub, O] for any operator O ∈ {M_z, P_z, L, S, Q, γ^5, P_op} from the operator zoo. Specializing O = H_sub gives dH_sub/dt = 0 (self-commutator vanishes); energy conservation is the trivial statement at operator level.

Proof sketch (3 ingredients):

1. SO_0(5,2) acts as holomorphic isometries on D_IV^5: by Cartan's theorem on bounded Hermitian symmetric domains, SO_0(5,2) is the full isometry group of D_IV^5 acting transitively. The action lifts to a unitary representation π on Bergman $H^2(D\_IV^5)$ (Wallach 1976).
2. Casimir commutes with all of so(5,2): by the standard Lie algebra fact, the universal Casimir of so(5,2) commutes with every element of the algebra (Casimir element is central in U(so(5,2))). In particular, [C_{so(5,2)}, T_E] = 0 where T_E is the time-translation generator.
3. H_sub = Casimir restriction: per Elie K52a S29, H_sub = restriction of universal Casimir to the L^2-section bundle. Therefore [H_sub, T_E] = 0, and by Noether's theorem energy E = ⟨H_sub⟩ is conserved.

Standard-model match: - Energy conservation observed at all scales tested (atomic clocks to cosmological); CONFIRMED - T2473 substrate-derivation matches standard QFT statement: time-translation symmetry → energy conservation - BST contribution: the specific Hamiltonian H_sub = Casimir on L^2(D_IV^5; L_λ) is forced, not postulated

Status: STRUCTURALLY VERIFIED candidate. Reduces to Wallach 1976 + Casimir centrality + Noether's theorem. Foundation for all energy-related observable derivations (atomic spectra, hadronic mass spectra, particle masses).

Cross-references: T2438 Elie H_sub framework, T2435 Casimir algebra, Vol 0 Ch 4 §4.6 SO_0(5,2) conformal structure, Vol 1 Ch 7 Schrödinger picture, Wallach 1976, Noether 1918.

Verification toy: Toy 3505 (`toy_3505_t2473_energy_conservation_substrate.py`, 8-test specification — pending Elie/cross-lane build): - (T1) SO_0(5,2) acts on D_IV^5 by holomorphic isometries - (T2) Lifted unitary representation π on Bergman H^2(D_IV^5) - (T3) Casimir C_{so(5,2)} commutes with all so(5,2) elements - (T4) Casimir-action eigenvalue = C_2 = 6 at ground state - (T5) H_sub time-translation invariance [H_sub, T_E] = 0 - (T6) Noether-conserved E = ⟨ψ|H_sub|ψ⟩ stable under substrate-tick evolution - (T7) Heisenberg dO/dt = (i/$\hbar$)[H_sub, O] gives dH_sub/dt = 0 - (T8) Wallach K-type Casimir spectrum exhausts H_sub eigenvalues

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T2474 — Momentum Conservation via Coset Translation-Direction Invariance

Statement: Let P_z = ∂_z (Wirtinger derivative, momentum operator T2422 RATIFIED) be the substrate momentum operator on Bergman H^2(D_IV^5). The substrate space-translation generators are the 5+5 = 10 coset directions in m ⊂ so(5,2)/(so(5)⊕so(2)) complement (Vol 0 Ch 4 §4.2 v0.5). Then:

(a) Translation symmetry: each of the 10 coset translation generators T_i commutes with H_sub at sector-restricted level (no external boundary forces): [T_i, H_sub] = 0.

(b) Noether conservation: by Noether's theorem applied to each T_i, the corresponding charges — the substrate momentum components ⟨ψ| P_z |ψ⟩ — are conserved under substrate-tick evolution in the absence of substrate-coupled boundary interactions.

(c) Vectorial structure: the 10 coset generators decompose under SO(5) × SO(2) into a 5-dim SO(5)-vector (spatial momenta P_x, P_y, P_z, P_4, P_5) + a 5-dim SO(5)-vector dual via SO(2) action. Restriction to 4D conformal boundary (Vol 0 Ch 4 §4.6) gives the standard 4-momentum (p_0, p_1, p_2, p_3) plus a 5th "internal" momentum direction.

(d) Operator level: dP_z/dt = (i/ħ)[H_sub, P_z] = 0 (by symmetry), giving conservation in Heisenberg picture. The canonical commutator [M_z, P_z] = −I (T2422 RATIFIED) is preserved under time evolution.

Proof sketch (3 ingredients):

1. Coset translation directions exist in m: the symmetric pair so(5,2) = (so(5) ⊕ so(2)) ⊕ m has 10-dimensional complement m (since dim so(5,2) = 21 and dim(so(5)⊕so(2)) = 11). The m direction generates translations on D_IV^5 via the exponential map at the base point.
2. Each m-generator commutes with Casimir: by the standard Lie algebra fact (Casimir central in U(so(5,2))), each m_i ∈ m commutes with the universal Casimir. Restricting to L^2-section bundle gives [H_sub, m_i] = 0.
3. Substrate momentum P_z is unitary representation of m_z: under the exponential map, the coset direction m_z generates translations e^{i t m_z} on Bergman $H^2(D\_IV^5)$. The infinitesimal generator IS the substrate momentum operator P_z. Hence dP_z/dt = (i/ħ)[H_sub, P_z] = 0.

Standard-model match: - Momentum conservation observed at all collision experiments; CONFIRMED - BST 4-momentum reduces to standard 4-momentum at conformal boundary (Vol 0 Ch 4 §4.6 SO_0(3,1) ⊂ SO_0(5,2)) - The 5th "internal momentum direction" corresponds to substrate-internal degree of freedom (related to N_max scale)

Status: STRUCTURALLY VERIFIED candidate. Reduces to Cartan symmetric-pair decomposition + Casimir centrality + Noether's theorem.

Cross-references: T2422 momentum operator, Vol 0 Ch 4 §4.2 v0.5 coset structure, T2438 + Elie K52a S29, Vol 1 Ch 7 dynamics, Wallach 1976, Noether 1918.

Verification toy: Toy 3506 (`toy_3506_t2474_momentum_conservation_substrate.py`, 8-test specification — pending build): - (T1) Cartan decomposition so(5,2) = (so(5)⊕so(2)) ⊕ m with dim m = 10 - (T2) Each m-generator m_i commutes with Casimir - (T3) Restriction to L^2-section: [H_sub, m_i] = 0 - (T4) Exponential map e^{i t m_z} generates translations - (T5) Infinitesimal generator = P_z (substrate momentum, T2422) - (T6) Noether-conserved ⟨ψ|P_z|ψ⟩ stable under substrate-tick evolution - (T7) [M_z, P_z] = −I canonical commutator preserved - (T8) Conformal-boundary 4-momentum reduction (5D internal → 4D + 1)

T2475 — Electric Charge Conservation via SO(2) Factor Invariance

Statement: Let Q = −i · dπ(J_{SO(2)}) be the substrate electric charge operator (T2470 STRUCTURALLY VERIFIED Friday). The **substrate U(1)_em ≅ SO(2) charge-rotation generator** is the SO(2) factor of the isotropy K = SO(5) × SO(2) acting on Bergman $H^2(D\_IV^5)$. Then:

(a) **U(1)_em symmetry**: in non-weak sectors (strong + EM Hamiltonians), the SO(2) factor of K commutes with H_sub|sector: [J_{SO(2)}, H_sub|strong+EM] = 0. (The weak sector breaks this via SU(2)_L × U(1)_Y → U(1)_em Weinberg mixing; conservation of Q post-Higgs-mechanism is preserved.)

(b) Noether conservation: by Noether's theorem applied to J_{SO(2)}, the corresponding charge — the total electric charge $\langle\psi|$ Q $|\psi\rangle$ — is conserved under substrate-tick evolution in all non-weak interactions, and conserved across all sectors post-electroweak-symmetry-breaking (Higgs mechanism leaves U(1)_em unbroken).

(c) Quantization: per T2470, Q spectrum is integers + {±1/N_c, ±2/N_c} = {±1/3, ±2/3} fractional. Total charge of any process is conserved + quantized.

(d) Operator level: dQ/dt = (i/$\hbar$)[H_sub, Q] = 0 (by symmetry, post-electroweak-breaking).

Proof sketch (3 ingredients):

1. SO(2) factor commutes with strong + EM Hamiltonians: by direct construction (Vol 0 Ch 4 §4.3), the SO(2) factor of K = SO(5) × SO(2) acts on substrate states as a global phase rotation. This phase is preserved by strong + EM Hamiltonians (which depend only on $|\psi|^2$, not phase). Hence [J_{SO(2)}, H_strong] = 0 + [J_{SO(2)}, H_EM] = 0.

2. Weak-sector breaking + Higgs mechanism: in the weak sector, the larger group SU(2)_L × U(1)_Y has SO(2) embedded in U(1)_Y; the Higgs mechanism breaks SU(2)_L × U(1)_Y → U(1)_em (unbroken). The unbroken U(1)_em is exactly the SO(2) substrate factor (with Weinberg-mixed phase). Conservation of Q post-Higgs is therefore preserved.

3. Substrate-derived total Q conservation: by T2470 + (1) + (2), the total Q over any process is conserved across all sectors.

Standard-model match: - Electric charge conservation observed at all scales; CONFIRMED - BST substrate-derivation matches standard QFT: U(1)_em symmetry → Q conservation - BST contribution: Q quantization is substrate-derived (T2470); integer + {±1/3, ±2/3} spectrum exhausts.

Status: STRUCTURALLY VERIFIED candidate. Reduces to T2470 + Vol 0 Ch 4 §4.3 SO(2) factor + Higgs mechanism U(1)_em unbroken. Closes electric charge conservation substrate-derivation.

Cross-references: T2470 charge operator, Vol 0 Ch 4 §4.3 + §4.5 SO(2) factor, Vol 1 Ch 8 SM gauge group + Higgs mechanism, Vol 2 Ch 9 Higgs sector (Elie PARTIAL DERIVED), Noether 1918.

Verification toy: Toy 3507 (`toy_3507_t2475_charge_conservation_substrate.py`, 8-test specification — pending build): - (T1) SO(2) factor of K commutes with strong + EM Hamiltonians - (T2) Higgs mechanism breaks SU(2)_L × U(1)_Y

→ U(1)_em unbroken - (T3) Substrate SO(2) factor ≡ U(1)_em post-Weinberg-mixing - (T4) [Q, H_sub] = 0 across all sectors (post-electroweak-breaking) - (T5) Noether-conserved $\langle\psi|Q|\psi\rangle$ stable under substrate-tick evolution - (T6) Q quantization: integers + {±1/3, ±2/3} - (T7) Charge conservation across particle-physics processes (decay, scattering, annihilation) - (T8) dQ/dt = 0 in Heisenberg picture

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Connection to Vol 0 Ch 8 Conservation Laws + Strong-Uniqueness C12

Vol 0 Ch 8 Conservation Laws (Keeper-lane chapter-grade narrative): T2473 + T2474 + T2475 directly contribute to the 15 substrate-derivation conservation theorems target in Ch 8. Energy + momentum + charge are the three foundational conservation laws of mechanics; remaining 12 conservation laws (angular momentum, hypercharge, lepton number, baryon number, parity, time-reversal, charge conjugation, CPT, color, weak isospin, gluon-number, etc.) follow the same pattern: substrate-symmetry generator → Noether conserved quantity.

Strong-Uniqueness C12 (RIGOROUSLY CLOSED via T2441 operator zoo ground state): T2473 + T2474 + T2475 strengthen C12 by extending the operator-Hamiltonian commutation pattern to multiple conserved-charge operators. The substrate's organization is even more constrained than C12 alone implies.

Vol 1 Ch 4 Discrete Symmetries (Lyra v0.4 + T2472 P_op): per T2472 parity violation in weak sector + T2475 charge conservation post-electroweak-symmetry-breaking, the substrate-derivation of weak-sector physics is now closed at conservation-law level. Parity violated; charge conserved post-Higgs.

Filing status

v0.1: Friday afternoon 2026-05-22 ~14:45 EDT — Lyra theorem-writing lane per Casey + Keeper 14:23 EDT board Item #3 SP-31 #279 priority. Three substrate-derivation conservation-law theorems T2473 + T2474 + T2475 STRUCTURALLY VERIFIED candidates filed; toys 3505 + 3506 + 3507 specs included (24 tests total, pending Elie/cross-lane build).

Pending Cal cold-read (queued): - Tier 1: T2473 + T2474 + T2475 derivation rigor + Noether-substrate equivalence + Cal #99 v0.3 framings check - Tier 2: standard-model match verification (atomic-scale energy/momentum conservation + charge quantization)

Pending Keeper K-audit: - K180 energy conservation T2473 - K181 momentum conservation T2474 - K182 charge conservation T2475

Cross-CI handoff: - Elie: Toys 3505 + 3506 + 3507 specs (24 tests, ~30 min build); Vol 0 Ch 8 absorption - Grace: catalog cross-references for T2473+T2474+T2475 +

conservation-law backbone for Vol 0 Ch 8 - Keeper: K180 + K181 + K182 K-audit pre-stages + Vol 0 Ch 8 chapter-grade narrative integration

Cross-volume integration plan: - Vol 0 Ch 8 (Conservation Laws) v0.5 — primary integration; T2473+T2474+T2475 as Sections 8.1+8.2+8.3 first three conservation theorems - Vol 1 Ch 7 (Dynamics) — strengthen dO/dt = (i/ℏ)[H_sub, O] absorption - Vol 2 Ch 9 (Higgs) — cross-link T2475 charge conservation post-electroweak-breaking

SP-31 #279 progress: 3 of ~12-15 substrate-derivation conservation theorems filed. Remaining (multi-week): angular momentum, hypercharge, lepton number, baryon number, parity (T2472 already covers), time-reversal (T2433 covers), charge conjugation (T2434 covers), CPT (K87 covers), color SU(3), weak isospin, gluon-number, chirality (T2471 covers).

— Lyra, SP-31 #279 per-conservation-law substrate-derivation theorems T2473 + T2474 + T2475 v0.1, Friday 2026-05-22 ~14:45 EDT